

The Symmetric Group and Shuffling Cards

Chen Jiayi

What is a permutation?

- A **permutation** of $\{1, 2, \dots, n\}$ is a bijection $\sigma : [n] \rightarrow [n]$ (a reordering of $1, \dots, n$).
- Common notations: $[\sigma(1), \sigma(2), \dots, \sigma(n)]$,
- **Operation:** composition $(\sigma \circ \tau)(i) = \sigma(\tau(i))$.
- a map from positions $\rightarrow$ labels (card numbers); $\sigma(i)$ is the label in position i .

i	1	2	3	4
$\sigma(i)$	3	1	2	4

Example above: card 3 goes to position 1, card 1 to 2, card 2 to 3, card 4 stays in 4.

The symmetric group S_n

- **Definition:** S_n is the set of *all* permutations of $[n] = \{1, \dots, n\}$.
- **Identity & inverse:** $\text{id}(i) = i$; every $\sigma \in S_n$ has an inverse σ^{-1} .
- **Size:** $|S_n| = n!$.
- **Shuffling view:** a shuffle is a random walk on S_n : apply a random permutation to the current deck.

Why is the uniform distribution stationary on S_n ?

Let $G = S_n$. Each step pick $\sigma \sim \mu$ and update $x \mapsto \sigma \circ x$.

$$P(x, y) = \Pr(\sigma \circ x = y) = \mu(x^{-1}y).$$

Uniform $\pi(x) = 1/|G|$ gives

$$(\pi P)(y) = \sum_{x \in G} \frac{1}{|G|} \mu(x^{-1}y) = \frac{1}{|G|} \sum_{z \in G} \mu(z) = \frac{1}{|G|} = \pi(y).$$

Convergence: if $\text{supp}(\mu)$ generates S_n and $\mu(\text{id}) > 0$, the chain is irreducible+aperiodic, so it converges to the (unique) uniform stationary law.

Random transpositions

Shuffle rule. At time t , pick two labels $L_t, R_t \in \{1, \dots, n\}$ independently and uniformly. If $L_t \neq R_t$, swap those two cards; if $L_t = R_t$, do nothing.

One-step distribution μ (probability of a single-step permutation σ):

$$\mu(\sigma) = \begin{cases} \frac{1}{n}, & \sigma = \text{id}, \\ \frac{2}{n^2}, & \sigma = (ij), \ i \neq j, \\ 0, & \text{otherwise.} \end{cases}$$

Irreducible & aperiodic. Transpositions generate $S_n \Rightarrow$ irreducible; $\mu(\text{id}) > 0$ gives a self-loop $\Rightarrow$ aperiodic.

Random Transpositions: lower bound on mixing time

Fixed point. For a permutation σ , an index i is a *fixed point* if $\sigma(i) = i$.

Untouched probability (exact). One step picks $L, R \stackrel{\text{i.i.d.}}{\sim} \text{Unif}([n])$. For a fixed label k ,

$$\Pr(k \text{ untouched in one step}) = \Pr(L \neq k \ \& \ R \neq k) = \left(1 - \frac{1}{n}\right)^2.$$

After t steps,

$$\Pr(k \text{ untouched up to } t) = \left(1 - \frac{1}{n}\right)^{2t} \simeq e^{-2t/n}.$$

Hence

$$\mathbb{E}[\# \text{untouched}] = n \left(1 - \frac{1}{n}\right)^{2t} \simeq n e^{-2t/n}.$$

Lower bound on mixing. If a card is untouched, it stays in its original position $\Rightarrow$ it is a fixed point. A uniform random permutation has $\mathbb{E}[\# \text{fixed points}] = 1$. Setting $n(1 - \frac{1}{n})^{2t} \approx 1$ gives

$$t \gtrsim \frac{n}{2} \log n \quad (\text{up to an additive } O(n)).$$

Proposition 8.4

Let $0 < \varepsilon < 1$. For the random transposition chain on an n -card deck,

$$t_{\text{mix}}(\varepsilon) \geq \frac{n-1}{2} \log\left(\frac{1-\varepsilon}{6n}\right).$$

Proof.

$$d(t) := \max_x \|P^t(x, \cdot) - \pi\|_{\text{TV}}, \quad t_{\text{mix}}(\varepsilon) := \min\{t : d(t) \leq \varepsilon\}.$$

Hence, to prove $t_{\text{mix}}(\varepsilon) \geq T$ it suffices to show that at time T the chain is still at least ε away from stationarity:

$$d(T) > \varepsilon \quad (\text{equivalently, } \exists x : \|P^T(x, \cdot) - \pi\|_{\text{TV}} > \varepsilon).$$

Random Transpositions: lower bound on mixing time

- R_{2t} : number of *untouched* cards after $2t$ label draws (equivalently, after t random transpositions).
- $F(\sigma) = \#\{i \in [n] : \sigma(i) = i\}$: number of fixed points of σ .

Proof

$$\mu := \mathbb{E}[R_{2t}] = n\left(1 - \frac{1}{n}\right)^{2t}, \quad \text{Var}(R_{2t}) \leq \mu.$$

Let $A = \{\sigma : F(\sigma) \geq \mu/2\}$. Under the uniform law π on S_n ,

$$\pi(A) = \Pr_{\sigma \sim \pi}(F(\sigma) \geq \mu/2) \leq \frac{2}{\mu} \quad (\text{Markov: } \Pr(X \geq a) \leq \mathbb{E}X/a).$$

Since $F(\sigma_t) \geq R_{2t}$,

$$P^t(\text{id}, A^c) \leq \Pr(R_{2t} \leq \mu/2) \leq \frac{4}{\mu} \quad (\text{Chebyshev: } \Pr(|X - \mathbb{E}X| \geq a) \leq \text{Var}(X)/a^2).$$

Random Transpositions: lower bound on mixing time

Proof(continued)

$$d(t) = \|P^t(\text{id}, \cdot) - \pi\|_{\text{TV}} \geq 1 - \frac{6}{\mu}.$$

To ensure $d(t) > \varepsilon$, it suffices that

$$\mu > \frac{6}{1 - \varepsilon}.$$

With $\mu = n(1 - \frac{1}{n})^{2t}$, this is equivalent to

$$\log\left(\frac{n(1 - \varepsilon)}{6}\right) > 2t \log\left(\frac{n}{n-1}\right).$$

Using $\log(1 + x) < x$ (take $x = \frac{1}{n-1}$) gives

$$\log\left(\frac{n(1 - \varepsilon)}{6}\right) \geq \frac{2t}{n-1}.$$

Proof(continued)

Thus, if

$$t \leq \frac{n-1}{2} \log\left(\frac{n(1-\varepsilon)}{6}\right),$$

then $d(t) > \varepsilon$, and hence

$$t_{\text{mix}}(\varepsilon) \geq \frac{n-1}{2} \log\left(\frac{n(1-\varepsilon)}{6}\right).$$

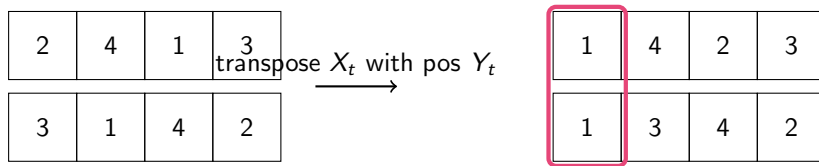
Upper bound on mixing time via coupling

We construct the same transition mechanism in a different way.

- At each time t , pick a **label** $X_t \in [n]$ and a **position** $Y_t \in [n]$ independently and uniformly.
- In a deck, *transpose* the card with label X_t with the card currently at position Y_t . If position Y_t already holds label X_t , do nothing.
- To **couple two decks**, use the *same* (X_t, Y_t) sequence for both decks.
- Let a_t be the number of positions where the two decks agree (same label in the same position). Then:
 - ① If label X_t is already in the same position in both decks, then $a_{t+1} = a_t$.
 - ② If X_t is in different positions but both decks have the *same* label at position Y_t , the step may break one agreement and create one; net effect $a_{t+1} = a_t$.
 - ③ If X_t is in different positions *and* the labels at position Y_t are different in the two decks, at least one new agreement is formed (sometimes up to three). Hence $a_{t+1} \geq a_t + 1$.
- So agreements never decrease and often increase; this drives the decks together.

Case 3: illustration

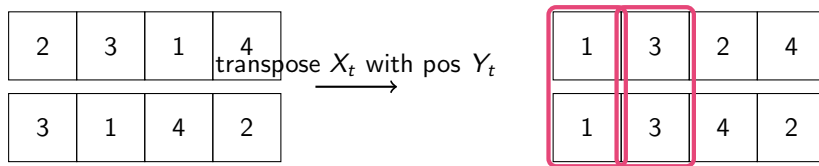
Aligning one card. (choose $X_t = 1$, $Y_t = 1$)



After the step, position 1 matches in both decks (highlighted).

Case 3: illustration

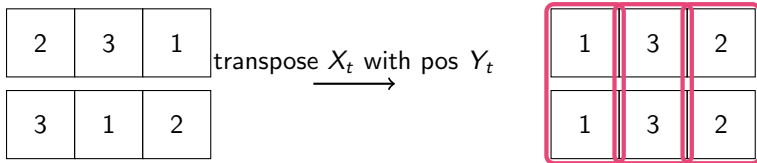
Aligning two cards. (again $X_t = 1$, $Y_t = 1$)



Now positions 1 and 2 match in both decks.

Case 3: illustration

Aligning three cards. (here $n = 3$, $X_t = 1$, $Y_t = 1$)



All three positions align after the step.

Proposition 8.5

Let τ be the time required for two decks to *coincide* (under the coupled random-transposition dynamics). Then, for every choice of initial configurations,

$$\mathbb{E} \tau < \frac{\pi^2}{6} n^2.$$

We couple two decks by using the same label/position sequence (X_t, Y_t) each step; let a_t be the number of agreements (same label in the same position) between the two decks.

Proof of Proposition 8.5

Proof

Level decomposition. Let a_t be the number of agreements between the coupled decks at time t . For $i = 0, 1, \dots, n-1$, define

$$\tau_{i+1} := \min\{t \geq 0 : a_t \geq i+1\} - \min\{t \geq 0 : a_t \geq i\}.$$

Then the coupling time decomposes as

$$\tau = \sum_{i=0}^{n-1} \tau_{i+1}.$$

One-step Success Probability.

$$p_i = \Pr(a_{t+1} = i+1 \mid a_t = i) = \frac{(n-i)^2}{n^2}.$$

$$\tau_{i+1} \mid (a_t = i) \sim \text{Geom}(p_i) \implies \mathbb{E}[\tau_{i+1} \mid a_t = i] = \frac{1}{p_i} = \frac{n^2}{(n-i)^2}.$$

Proof of Proposition 8.5

Proof(Continued)

$$\mathbb{E}\tau = \sum_{i=0}^{n-1} \mathbb{E}\tau_{i+1} < n^2 \sum_{i=0}^{n-1} \frac{1}{(n-i)^2} = n^2 \sum_{\ell=1}^n \frac{1}{\ell^2} < n^2 \sum_{\ell=1}^{\infty} \frac{1}{\ell^2} = \frac{\pi^2}{6} n^2.$$

$O(n^2)$ mixing time upper bound

Let τ be the coupling (coalescence) time for the standard coupling of random transpositions. Then

$$d(t) = \|P^t(\text{id}, \cdot) - \pi\|_{\text{TV}} \leq \Pr(\tau > t) \leq \frac{\mathbb{E}[\tau]}{t} \quad (\text{Markov's inequality}).$$

From Proposition 8.5, $\mathbb{E}[\tau] < \frac{\pi^2}{6} n^2$. Hence

$$d(t) < \frac{\pi^2}{6} \frac{n^2}{t}.$$

In particular, to ensure $d(t) \leq \varepsilon$ it suffices to take

$$t \geq \frac{\pi^2}{6\varepsilon} n^2 \implies t_{\text{mix}}(\varepsilon) \leq \frac{\pi^2}{6\varepsilon} n^2.$$

Remark. This is a crude $O(n^2)$ upper bound; the sharp order for random transpositions is $O(n \log n)$.

Upper bound on Mixing time via strong stationary time

Proposition 8.6

In the random-transposition shuffle, at step t choose two cards (hands) $L_t, R_t \in [n]$ independently and uniformly, and transpose them.

Start with no marks. At time t , *mark* R_t if both hold:

- R_t is currently unmarked; and
- either L_t is already marked, or $L_t = R_t$.

Let τ be the first time when every card is marked. Then τ is a *strong stationary time* for this chain (i.e., the deck is uniform at time τ , regardless of the past).

Consequence: $\|P^t(\text{id}, \cdot) - \pi\|_{\text{TV}} \leq \Pr(\tau > t).$

Proof sketch of Proposition 8.6

Let $V_t \subseteq [n]$ be the set of cards marked by time t , and $U_t \subseteq [n]$ the set of positions occupied by V_t after step t .

Claim. *Given t , V_t and U_t , all permutations of the cards V_t over the positions U_t are equally likely.*

Base $t = 1$: Trivial.

Step $t \rightarrow t + 1$: Choose L_{t+1}, R_{t+1} .

- ① *No new card is marked.* Three subcases (both unmarked, both marked, or R_{t+1} marked earlier and L_{t+1} unmarked). In each, the action on the marked block is either identity or a uniform transposition within U_t , so equiprobability on V_t over U_t is preserved.
- ② *R_{t+1} is newly marked.* Then $V_{t+1} = V_t \cup \{R_{t+1}\}$ and $U_{t+1} = U_t \cup \{\text{pos}(R_{t+1} \text{ at time } t)\}$. The new marked card and its position are each equally likely among the available choices, so *all* permutations of V_{t+1} over U_{t+1} are equally likely.

At time τ we have $V_\tau = [n]$ and $U_\tau = [n]$, hence the deck is uniform on S_n . Therefore τ is a strong stationary time and $\|P^t(\text{id}, \cdot) - \pi\|_{\text{TV}} \leq \Pr(\tau > t)$.

Lemma 8.9

Let τ be the stopping time from Proposition 8.6 (the time when every card is marked under the marking rule). Then

$$\mathbb{E} \tau = 2n(\log n + O(1)), \quad \text{and} \quad \text{Var}(\tau) = O(n^2).$$

Lemma 8.9: $\mathbb{E}\tau = 2n(\log n + O(1))$

Proof

Under the marking rule (Prop. 8.6),

$$\tau = \tau_0 + \tau_1 + \cdots + \tau_{n-1},$$

where τ_k is the number of steps after the k -th card is marked until the $(k+1)$ -st card is marked.

One-Step Success Probability.

$$p_k = \Pr(a_{t+1} = k+1 \mid a_t = k) = \frac{(k+1)(n-k)}{n^2}.$$

$$\tau_k \mid (a_t = k) \sim \text{Geom}(p_k) \quad \implies \quad \mathbb{E}[\tau_k \mid a_t = k] = \frac{1}{p_k} = \frac{n^2}{(k+1)(n-k)}.$$

Hence

$$\mathbb{E}\tau = \sum_{k=0}^{n-1} \frac{1}{p_k} = n^2 \sum_{k=0}^{n-1} \frac{1}{(k+1)(n-k)}.$$

Proof (Continued)

$$\frac{1}{(k+1)(n-k)} = \frac{1}{n+1} \left(\frac{1}{k+1} + \frac{1}{n-k} \right), \quad \sum_{j=1}^n \frac{1}{j} = \log n + O(1).$$

Therefore

$$\mathbb{E}_T = n^2 \cdot \frac{2}{n+1} (\log n + O(1)) = 2n(\log n + O(1)).$$

Lemma 8.9: $\text{Var}(\tau) = O(n^2)$

Proof

$$\text{Var}(\tau) = \sum_{k=0}^{n-1} \frac{1 - \frac{(k+1)(n-k)}{n^2}}{\left(\frac{(k+1)(n-k)}{n^2}\right)^2} < \sum_{k=0}^{n-1} \frac{n^4}{(k+1)^2(n-k)^2}.$$

$$\begin{aligned} \text{Var}(\tau) &< \sum_{0 \leq k < n/2} \frac{n^4}{(k+1)^2(n-k)^2} + \sum_{n/2 \leq k < n} \frac{n^4}{(k+1)^2(n-k)^2} \\ &< \frac{2n^4}{(n/2)^2} \sum_{0 \leq k \leq n/2} \frac{1}{(k+1)^2} = O(n^2). \end{aligned}$$

Upper bound via strong stationary time

Corollary 8.10.

For the random-transposition chain on an n -card deck,

$$t_{\text{mix}} \leq (2 + o(1)) n \log n.$$

Proof.

Let τ be the strong stationary time from Proposition 8.6, and set

$$t_0 = \mathbb{E}[\tau] + 2\sqrt{\text{Var}(\tau)}.$$

By Chebyshev's inequality,

$$\Pr\{\tau > t_0\} \leq \frac{\text{Var}(\tau)}{(2\sqrt{\text{Var}(\tau)})^2} = \frac{1}{4}.$$

Upper bound via stationary time

Proof.

By the strong-stationary-time inequality (Prop. 6.11),

$$d(t) = \|P^t(\text{id}, \cdot) - \pi\|_{\text{TV}} \leq \Pr\{\tau > t\},$$

Hence, $t_{\text{mix}}(1/4) \leq t_0$, and $t_{\text{mix}} \leq (2 + o(1)) n \log n$



Riffle Shuffles — Three Equivalent Models

1) Split & Interleave (positions)

- Choose $M \sim \text{Binom}(n, \frac{1}{2})$; split deck into top M and bottom $n - M$.
- Interleave uniformly among $\binom{n}{M}$ ways, preserving order within each pile.

2) Drop-by-Drop (probabilistic merge)

- After the split, hold piles over the table and drop cards one at a time.
- If left has a cards and right has b , next card is from left with prob. $a/(a + b)$ (right: $b/(a + b)$).

3) Label & Stable Sort (inverse riffle)

- Give each card an independent fair bit (0/1).
- Stable sort: all 0s first (relative order kept), then all 1s.

A **rising sequence** is a longest possible block of consecutive labels that appear in the right order when you look through the deck; they don't have to be next to each other, just to the right of the previous one.

Distribution produced by Methods (1) and (2)

Distribution Q on permutations $\sigma \in S_n$:

$$Q(\sigma) = \begin{cases} \frac{n+1}{2^n}, & \sigma = \text{id}, \\ \frac{1}{2^n}, & \sigma \text{ has exactly two rising sequences,} \\ 0, & \text{otherwise.} \end{cases}$$

Reversal: Method (3) generates $\hat{Q}$

Reverse distribution: For any distribution R on S_n , $\hat{R}(\rho) = R(\rho^{-1})$.

Method (3) (label & stable sort) gives $\hat{Q}$:

- Label cards independently with bits 0/1; in ρ^{-1} , all 0-labeled cards form one increasing sequence and all 1-labeled cards form the other.
- id arises from the $n+1$ strings $00 \dots 011 \dots 1$.
- $M = \#\{0s\} \sim \text{Binom}(n, \frac{1}{2})$; given M , the positions of the 0s are uniform among $\binom{n}{M}$ choices.
- Hence Method (3) is the time-reversal of Method (1), i.e. it generates $\hat{Q}$.

Example: Inverse Riffle Shuffle

Initial order:

1 2 3 4 5 6 7 8 9 10 11 12 13

Round 1 labels: 1 0 0 1 1 1 0 1 0 1 1 0 0

Round 2 labels: 0 1 0 1 0 1 1 1 0 0 1 0 1

After one shuffle: 2 3 7 9 12 13 1 4 5 6 8 10 11

After two shuffles: 3 9 12 1 5 10 2 7 13 4 6 8 11

Proposition 8.11.

Let τ be the number of inverse riffle shuffles required for all cards to have different bitstring labels. Then τ is a strong stationary time.

Proof. Condition on $\tau = t$. Each card receives a random bitstring of length t from fair coin flips. Since all bitstrings are distinct, and assigned exchangeably to the cards, the permutation is fully determined by the labels and is *uniform*, regardless of t .

Riffle Shuffle: Upper Bound on Mixing time

Proposition 8.12.

For the riffle shuffle on an n -card deck,

$$t_{\text{mix}}(1/4) \leq 2 \log_2 \left(\frac{4n}{3} \right) \quad \text{for sufficiently large } n.$$

Proof.

Consider the *inverse* riffle shuffle and let τ be the first time all n cards have distinct t -bit labels. Then τ is a strong stationary time, so $d(t) \leq \Pr(\tau > t)$.

Success probability after t rounds.

$$\Pr(\tau \leq t) = \prod_{k=0}^{n-1} \left(1 - \frac{k}{2^t} \right).$$

Riffle Shuffle: Upper Bound on Mixing time

Proof(Continued)

With $t = 2 \log_2(n/c)$ (so $2^t = n^2/c^2$),

$$\log \Pr(\tau \leq t) = -\frac{c^2}{2} + O\left(\frac{1}{n}\right) \Rightarrow \Pr(\tau \leq t) \approx e^{-c^2/2}.$$

Choose $c < \sqrt{2 \log(4/3)}$ (e.g. $c = \frac{3}{4}$) so that $\Pr(\tau > t) \leq \frac{1}{4}$. Hence

$$t_{\text{mix}}(1/4) \leq 2 \log_2\left(\frac{n}{c}\right) \leq 2 \log_2\left(\frac{4n}{3}\right).$$

Riffle Shuffle: Lower Bound

Proposition 8.13.

Fix $0 < \varepsilon, \delta < 1$. Consider riffle shuffling an n -card deck. For sufficiently large n ,

$$t_{\text{mix}}(\varepsilon) \geq (1 - \delta) \log_2 n.$$

Proof.

In one step of the inverse riffle, a move is determined by n independent fair bits, so from any state there are at most 2^n accessible next states. Hence the maximum out-degree satisfies $\log_2 \Delta \leq n$.

The state space has size $|\Omega| = n!$. By Stirling,

$$\log_2(n!) = [1 + o(1)] n \log_2 n.$$

Proof(Continued)

Apply the general counting bound (eq. (7.2)):

$$t_{\text{mix}}(\varepsilon) \geq \frac{\log_2((1-\varepsilon)|\Omega|)}{\log_2 \Delta} \geq \frac{\log_2(n!) + O(1)}{n} = \log_2 n - \log_2 e + o(1).$$

Given any fixed $\delta \in (0, 1)$, the additive constant is negligible for large n , so

$t_{\text{mix}}(\varepsilon) \geq (1 - \delta) \log_2 n$ for sufficiently large n .